

Global Retail Sales Analysis & Executive Dashboard Project

1. Project Overview

This project analyzes global retail sales data to identify performance trends, profitability drivers, and operational inefficiencies. The analysis focuses on sales, profit margins, product categories, discount impact, and regional performance.

The objective is to transform raw transactional data into actionable insights using data cleaning, exploratory analysis, and interactive dashboards.

The final output is an executive Tableau dashboard that enables stakeholders to quickly understand sales performance and make informed business decisions.

2. Business Problem

Retail companies generate large volumes of transactional data but often struggle to convert it into actionable insights.

Key business questions addressed:

- Which product categories generate the highest revenue?
- Which regions contribute the most to company profits?
- How do discounts affect profitability?
- Which products generate losses?
- Which customers generate the most value?

3. Data Source

The dataset contains global retail sales transactions.

Key fields include:

- Order ID
- Order Date
- Ship Date

- Region
- Country
- State
- City
- Category
- Sub-category
- Product Name
- Sales
- Quantity
- Discount
- Profit
- Customer ID
- Customer Name
- Segment
- Shipping Mode

4. Data Cleaning & Preparation

Key data preparation steps:

- Identified missing values using COUNTBLANK
- Standardized text using TRIM and PROPER functions
- Converted currency values using exchange rates
- Calculated Profit Margin = Profit / Sales
- Calculated Discount Amount = Sales × Discount
- Used lookup tables to map states to regions

5. Exploratory Data Analysis

Pivot tables were used to analyze:

- Sales by category
- Profit by product
- Regional performance
- Discount vs profit relationships

Scatter charts helped evaluate the relationship between discounts and profitability.

6. Customer Analysis (RFM Model)

RFM analysis was used to segment customers based on purchasing behavior.

Recency: Days since last purchase

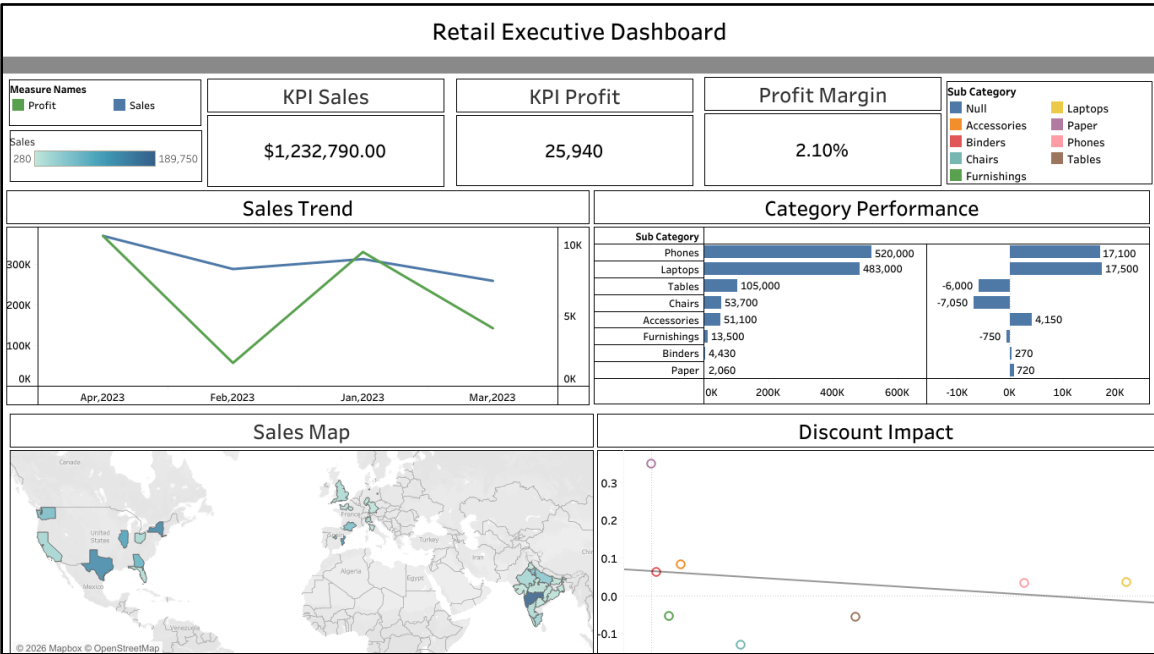
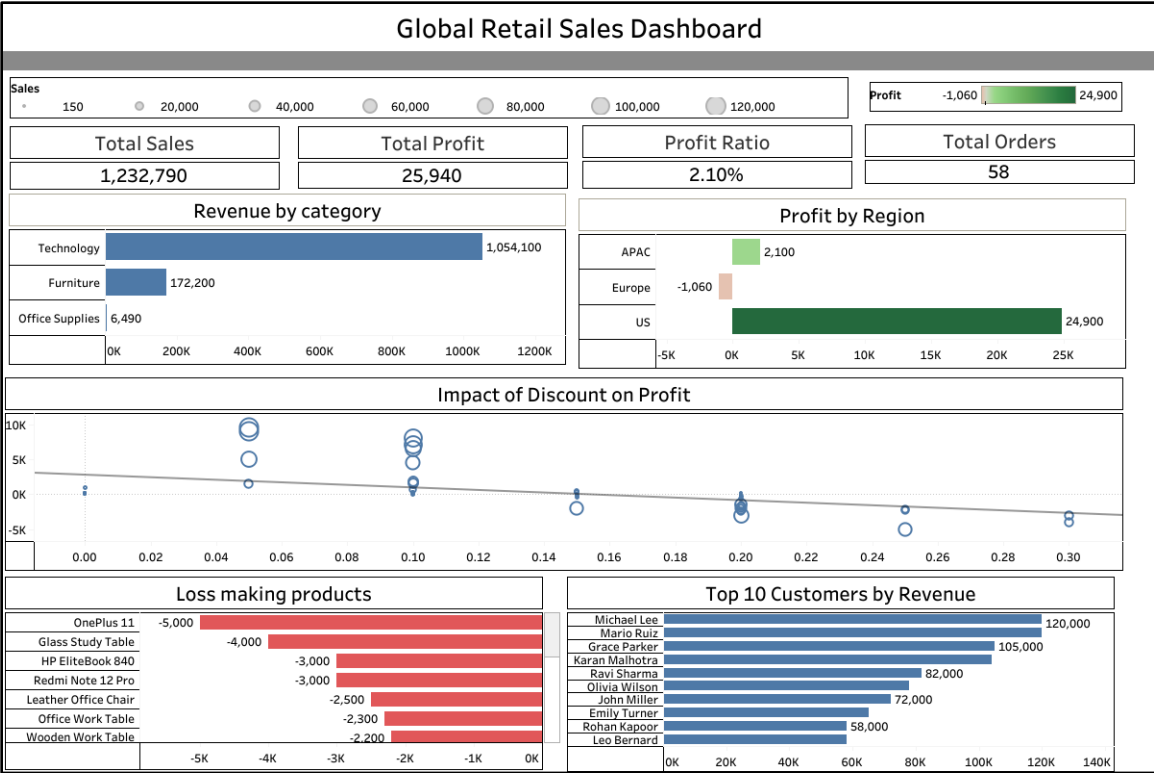
Frequency: Number of purchases

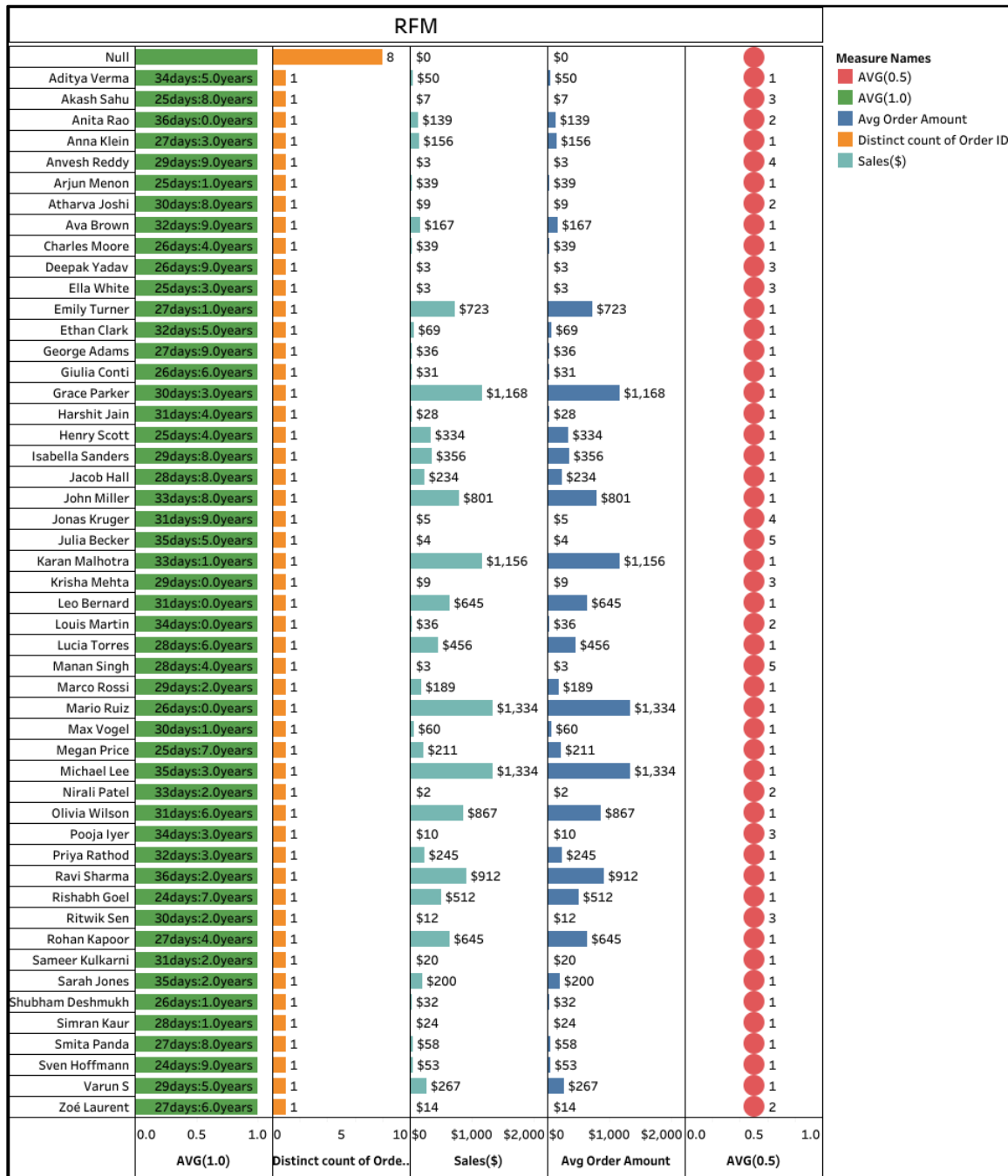
Monetary: Total revenue generated by customer

This helps identify high-value customers and customer segments.

7. Tools & Technologies

- Excel – Data cleaning and pivot table analysis
- Tableau – Data visualization and dashboard creation





8. Key Metrics

- Sales Revenue
- Profit
- Profit Margin

- Discount Impact
- Year-over-Year Sales Growth

9. Dashboard Components

Executive dashboard includes:

- KPI cards for sales, profit, and profit margin
- Monthly sales trend
- Geographic sales map
- Category performance chart
- Discount impact scatter plot
- Customer RFM analysis

10. Key Insights

- Higher discounts significantly reduce profit margins
- Some categories drive a majority of revenue
- Regional sales performance varies significantly
- Some products generate consistent losses
- A small segment of customers generates most revenue

11. Business Recommendations

- Optimize discount strategy
- Focus marketing on high-performing categories
- Improve operations in underperforming regions
- Review or discontinue loss-making products
- Implement loyalty programs for high-value customers

12. Skills Demonstrated

- Data Cleaning
- Exploratory Data Analysis

- Business Intelligence
- Customer Segmentation (RFM)
- Data Visualization
- Dashboard Design